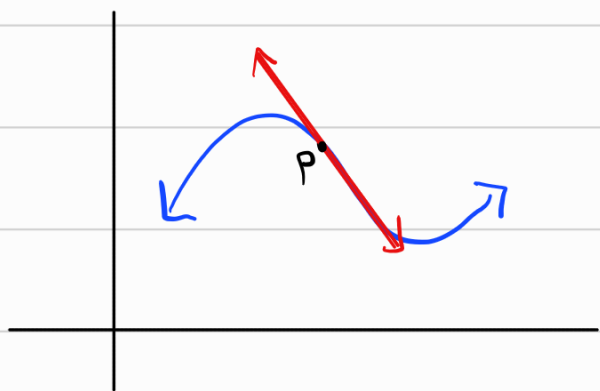


The Tangent Problem:

A tangent line is a line that intersects a curve (locally) one time.



These lines are unique as their slope is determined entirely by the function in question.

How do we find slope?

The issue is how we define slope $\left(\frac{\Delta y}{\Delta x}\right)$ requires two points, but for a tangent line we only have one point.

To fix this issue, we look at secant lines.



As Q moves closer to P , the secant line becomes a much better approximation of the tangent line.

Since a secant line has two intersection points, we can find slope!

In general, we use the following formula (difference quotient)

$$\frac{f(x) - f(a)}{x - a}$$

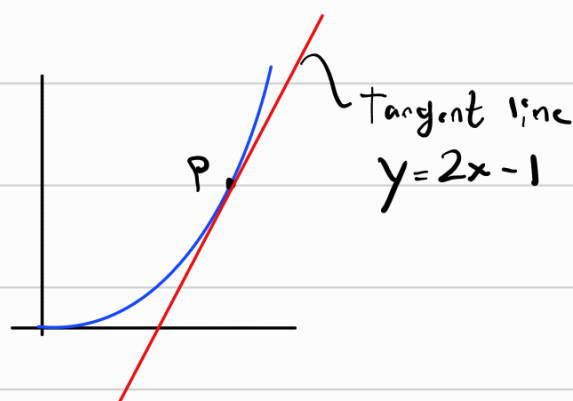
where a is the point $P = (a, f(a))$ and x is a close point to a .

Example:

$$f(x) = x^2 \text{ at } a=1$$

$$m_{PQ} = \frac{f(x) - f(1)}{x - 1} = \frac{x^2 - 1}{x - 1} \underset{\substack{= x+1 \\ x \neq 1}}{=} \frac{x^2 - 1}{x - 1}$$

x	m_{PQ}
2	3
1.5	2.5
1.25	2.25
1.01	2.01



$\vdots$
 $m = 2 \leadsto$ called a limit $\lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1} = 2$

The idea of a limit is discussed later, but it just considers "what happens as we get extremely close to a point but not actually equal to that point".

A question that often comes up is "why do we care about tangent lines?"

The reason is that they tell us instantaneous rate of change. So if a function f gives position over time, the slope of the tangent line at P tells the instant velocity at P (since distance/time is velocity). This works for any units, we mainly care about velocity and acceleration.

The Velocity Problem:

Consider a ball dropped from a high building.

We can measure the distance it has traveled over time with

$$S(t) = 4.9t^2 \text{ meters, } t\text{-seconds}$$

↳ Credit to Galileo

Using the ideas of tangent lines, we can estimate the velocity after, say, 5 seconds

$$\text{Avg Velocity} = \frac{\text{Change in Pos.}^{\text{tion}}}{\text{Change in time}}$$

$$= \frac{s(x) - s(5)}{x - 5} = \frac{s(5+h) - s(5)}{h}$$

Interval	x	V _{avg}
[5, 5.1]	5.1	49.49
[5, 5.01]	5.01	49.049
[5, 5.001]	5.001	49.0049
[5, 5.0001]	5.0001	49.00049

$$V_{\text{int}} = \lim_{x \rightarrow 5} \frac{s(x) - s(5)}{x - 5} = \boxed{49 \text{ m/s}} \quad \text{Velocity after 5 seconds}$$